RBE101TC: Engineering Mathematics

1st SEMESTER 2026–27 Coursework 1

DEPARTMENT: School of Robotics

Q1.

Let $z = -1 + i$.

(i) The principal argument $\mathrm{Arg}(z)$ is _____

(ii) The algebraic form of z^4 is _____

Q2.

Let C be the positively oriented circle $|z| = 2$. Evaluate $I = \oint_C \frac{e^z}{z-1}\, dz$.

The singularity of the integrand at $z = 1$ is _____

A) Removable singularity
B) Simple pole
C) Essential singularity
D) Pole of order 2

The value of the integral I is _____

Q3.

Let $f(z) = u(x,y) + iv(x,y)$ be analytic, where $u(x,y) = x^2 - y^2$.

From the Cauchy–Riemann equations, v_x equals _____

The corresponding analytic function $f(z)$ (with $v(0,0) = 0$) is _____

Q4.

Consider the power series $\sum_{n=0}^{\infty} \frac{(3n)!}{(n!)^3}(z+2-i)^n$.

The center z_0 of the series is _____

Its radius of convergence is _____

Q5.

Expand $f(z) = \frac{1}{z+3}$ about the center $z_0 = 0$.

The Taylor series valid in the region $|z| < 3$ is _____

The Laurent series valid in the region $|z| > 3$ is _____

Q6.

Let $f(z) = \frac{7z+1}{(z-2)(z+1)^3}$.

The singularity at $z = -1$ is _____

A) Removable singularity
B) Simple pole
C) Pole of order 2
D) Pole of order 3

The residue $\text{Res}_{z=-1} f(z)$ is _____

Q7.

Evaluate $\int_{-\infty}^{\infty} \frac{1}{x^2+4}\, dx$.

The pole of $f(z) = \frac{1}{z^2+4}$ lying in the upper half-plane is _____

The value of the integral is _____

Q8.

Consider $f(z) = e^{1/z}$.

The singularity at $z = 0$ is _____

A) Removable singularity
B) Simple pole
C) Essential singularity
D) Pole of order 2

The residue $\text{Res}_{z=0} f(z)$ is _____

Q9.

Consider the function

$$f(z) = \frac{e^{iz}}{z^2 + 1}.$$

(i) The singularities of $f(z)$ are_____

A) $z = \pm 1$

B) $z = \pm i$

C) $z = 0$

D) $z = \pm i$and $z = 0$

(ii) The residue of $f(z)$at the singularity lying in the upper half-plane is _____

A) $\frac{e^{-1}}{2i}$

B) $\frac{e^{-1}}{2}$

C) $\frac{e^{1}}{2i}$

D) $\frac{e^{1}}{2}$

Q10.

Evaluate $I = \int_0^{2\pi} \frac{d\theta}{5 - 4\cos\theta}$.

Using the substitution $z = e^{i\theta}$, $\cos\theta$ is expressed as _____

Given that $z = \frac{1}{2}$ is inside the unit circle with residue $\frac{1}{3}$, the integral equals _____